

ONLINE TEST:

Do refer Girish's experience regarding the online assessment(Everyone had the same questions).

The interviews took place at CUIC , and the interviewers came in person so we were not required to bring our laptops. We wrote the code in paper.

Technical interview 1:

1. Given a array , check if the array elements can be made consecutive or not.
Example : For array [5 , 2 , 1 , 3 , 4] our function must return the answer as true and for [1 , 2 , 3 , 3 , 4,5] we must return false (all the elements of the array must be included while making them consecutive) .
Initially the interviewer asked me what is the easiest approach you could think of , for which I said we can sort the array and then check if every element is one more than the previous element. He was satisfied with the answer and asked me to write the code for it. Then he asked about the time and space complexity of the code. And asked me to reduce the time complexity of the code. So I proposed an approach using hashmap and was again asked to write the code for it and we discussed about the time and space complexity of the code. Then he asked for a approach that is $O(n)$ time and $O(1)$ space complexity. I asked if we can use the formula for the sum of first n natural numbers , the interviewer was surprised (no idea why) and modified the problem saying the array can have negative values and zeroes. So after a few seconds I gave an approach where we modify the input array (Place the minimum element in the 0^{th} index , $\text{min} + 1$ in the 1^{st} index and so on) . He asked how this was different from sorting . In sorting we don't know the position where an element would end up , it is relative to other elements whereas in this method I know the position to place an element. Then I was asked to write the code for this , I had some difficulty writing the code and the interviewer helped me a lot while writing the code.

The question link

<https://www.geeksforgeeks.org/check-array-elements-consecutive-time-o1-space-handles-positive-negative-numbers/>

A leetcode question on which my last approach was based (modifying the input array)

<https://leetcode.com/problems/first-missing-positive/>

2. The second question was very easy compared to the first one.
<https://www.geeksforgeeks.org/count-1s-sorted-binary-array/>

I was not asked to write the code for this, the interviewer just asked of the various approaches I could think of. He was not interested in binary search $O(\log n)$ or iterating through the array and counting the number of one's. He was satisfied only after, I said that the sum of the array will be equal to the number of one's in it.

This round went on for 45 minutes or so and I was immediately asked to shift to the next room for round two.

Technical Round 2:

The interviewer started off with my DBMS project. I was asked to draw the database tables used in our project and draw the relationships among them, and write the primary key for each table. I hadn't normalised the tables in my DBMS project, and there were some redundant data in the tables. He asked on how to reduce the redundancy and gave me some advices regarding normalising tables. Then we moved on to DSA questions.

1. Check if a linked list is circular or not.
2. For this I was not given the direct question and was instead given a sample input and output and was asked to guess what the question might be. The first question was some modification of twoSum problem. But I was asked to write code for regular two sum , for which I wrote the brute , time optimized(using hashmap) and space optimized(sorting) solutions.
3. Again for the third question a sample input and output were given. The question was given two arrays find out what element to swap between them so that both the arrays have the same sum. I took a long time to find out the question from the sample input and outputs so the interviewer himself said what the question was. I wrote the brute force approach for the problem and was just asked on how to optimise it .

This round went on for an hour.

HR ROUND

The HR round was very short and nothing much happened there. I was asked to introduce myself , my family background and my interests. The one question which I think was important was “How do you rate yourself for the first two technical rounds”.

Key Takeaways

- Practice DSA regularly, refer ACM Code questions they are really helpful and practice leetcode and interviewbit.
- Prepare your projects before appearing for the interview, especially your DBMS project. Develop your project with the utmost importance to the backend.
- Never get disheartened if you were not selected for an interview (the online coding assessment)
- Have a good grasp of the core concepts of DBMS and OS.

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